

DBMS

Explain DBMS.

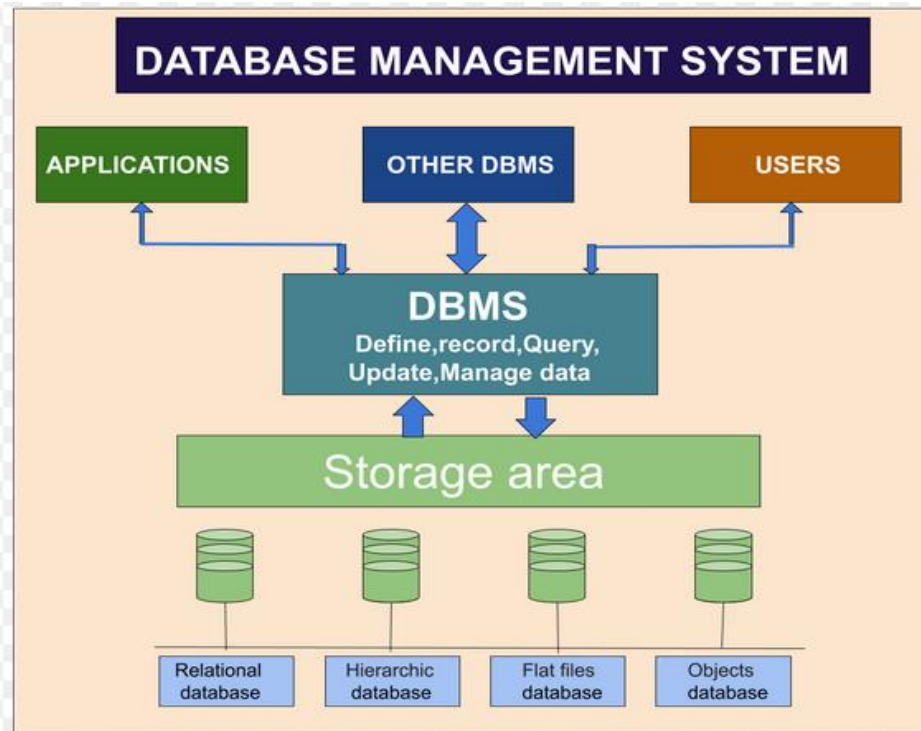
DBMS stands for database management systems. It is a collection of technical schemas that build and sustain a database. The database management system is software that provides a systematic method to interact with applications and databases to examine and produce the required outcome of the data. The database management system offers several operations. These include data insertion, data deletion, database creation, modification, and many more.

The top features of DBMS include the following-

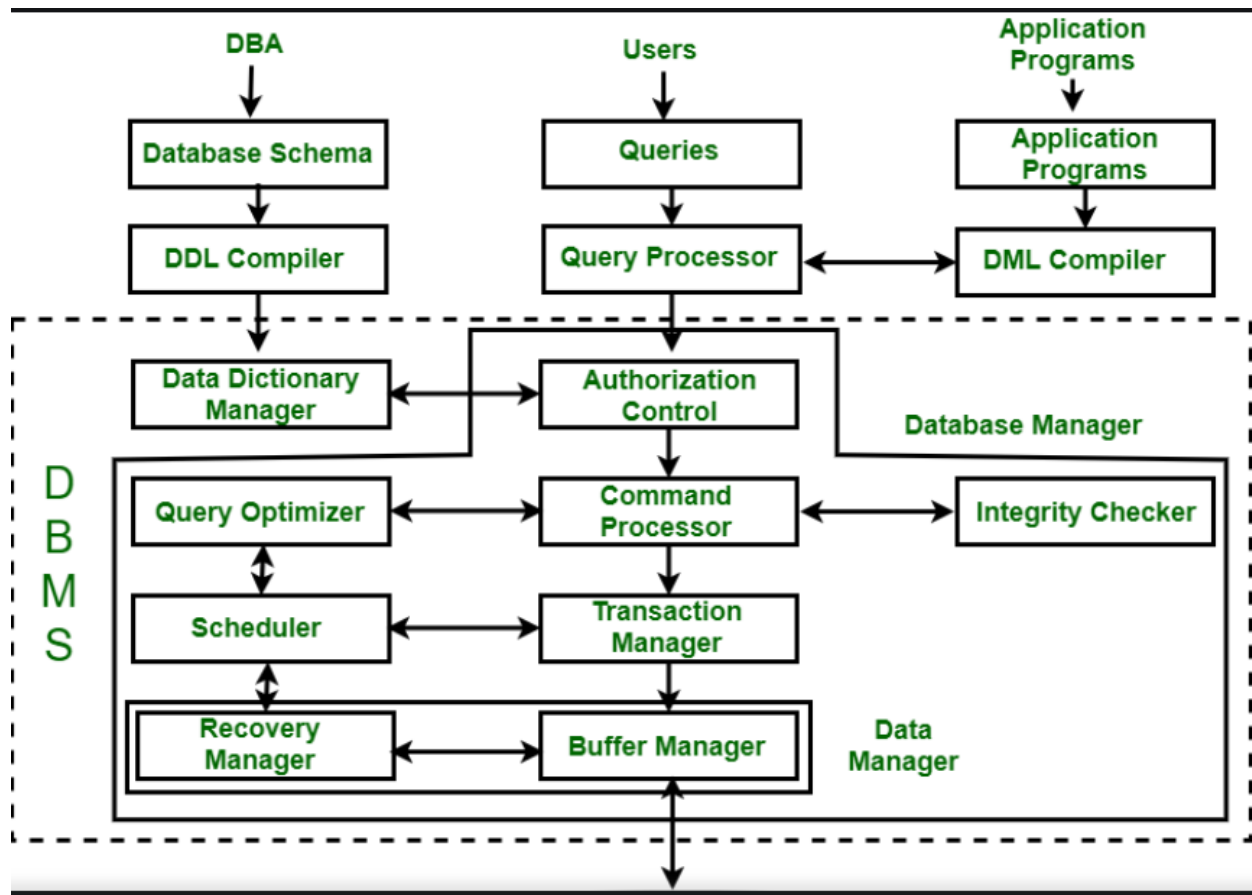
- Removes redundancy
- Large Database Maintenance
- It provides high security
- Saves storage
- Cost-effective
- The SQL language is used to insert, retrieve or remove any type of data item.
- Data integrity
- Supports multiple views of data
- Allows entities and relations among them to form tables.
- Follows ACID (Atomicity, Consistency, Isolation, Durability.) concept
- Supports multi-user environment

Advantages of using DBMS-

- Data consistency
- Data Integrity
- Data backups
- Applications customisation
- Data Management
- Data durability



Structure of Database Management System



Architecture of DBMS

1. Query Processor: It interprets the requests (queries) received from end user via an application program into instructions. It also executes the user request which is received from the DML compiler.

Query Processor contains the following components –

- **DML Compiler:** It processes the DML statements into low level instruction (machine language), so that they can be executed.
- **DDL Interpreter:** It processes the DDL statements into a set of table containing meta data (data about data).
- **Embedded DML Pre-compiler:** It processes DML statements embedded in an application program into procedural calls.
- **Query Optimizer:** It executes the instruction generated by DML Compiler.

2. Storage Manager: Storage Manager is a program that provides an interface between the data stored in the database and the queries received. It is also known as Database Control System. It maintains the consistency and integrity of the database by applying the constraints and executing the [DCL](#) statements. It is responsible for updating, storing, deleting, and retrieving data in the database.

It contains the following components –

- **Authorization Manager:** It ensures role-based access control, i.e., checks whether the particular person is privileged to perform the requested operation or not.
- **Integrity Manager:** It checks the integrity constraints when the database is modified.
- **Transaction Manager:** It controls concurrent access by performing the operations in a scheduled way that it receives the transaction. Thus, it ensures that the database remains in the consistent state before and after the execution of a transaction.
- **File Manager:** It manages the file space and the data structure used to represent information in the database.
- **Buffer Manager:** It is responsible for cache memory and the transfer of data between the secondary storage and main memory.

3. Disk Storage: It contains the following components –

- **Data Files:** It stores the data.
- **Data Dictionary:** It contains the information about the structure of any database object. It is the repository of information that governs the metadata.
- **Indices:** It provides faster retrieval of data item.

The structure of a Database Management System (DBMS) can be divided into three main components: The Internal Level, the Conceptual Level, and the External Level.

1. **Internal Level:** This level represents the physical storage of data in the database. It is responsible for storing and retrieving data from the storage devices, such as hard drives or solid-state drives. It deals with low-level implementation details such as data compression, indexing, and storage allocation.
2. **Conceptual Level:** This level represents the logical view of the database. It deals with the overall organization of data in the database and the relationships between them. It defines the data schema, which includes tables, attributes, and their relationships. The conceptual level is independent of any specific DBMS and can be implemented using different DBMSs.
3. **External Level:** This level represents the user's view of the database. It deals with how users access the data in the database. It allows users to view data in a way that makes sense to them, without worrying about the underlying implementation details. The external level provides a set of views or interfaces to the database, which are tailored to meet the needs of specific user groups.

The three levels are connected through a schema mapping process that translates data from one level to another. The schema mapping process ensures that changes made at one level are reflected in the other levels.

In addition to these three levels, a DBMS also includes a Database Administrator (DBA) component, which is responsible for managing the database system. The DBA is responsible for tasks such as database design, security management, backup and recovery, and performance tuning.

Overall, the structure of a DBMS is designed to provide a high level of abstraction to users, while still allowing low-level implementation details to be managed effectively. This allows users to focus on the logical organization of data in the database, without worrying about the physical storage or implementation details.

Question 2: What Are the Advantages of a Database Management System?

The significant benefits of the database management system are listed below:

Consistency: Data is duplicated in the traditional file system. Hence, any modifications made in one part may reflect on all other parts. This further leads to inconsistent information. We need to eliminate duplicate data to remove inconsistent information throughout the system. This is possible with a DBMS.

Elimination of Data Redundancy: The major reason DBMS is favoured over the traditional file system is that it controls data redundancy. Each user handles their file in the conventional file systems. This further leads to redundancy of the same information in several distinct files, errors, wastage of resources, and storage space.

Better Integrity: There is a lack of integrity in traditional file systems. In the DBMS

approach, it is crucial to impose integrity restrictions. The database approach is much better in terms of integrity than the conventional file system because it is centralized. In the database management system, data can be utilized by several users at a time.

Requirements Can be Examined: With the implementation of a centralized data approach, it is much easier to analyse organizations' needs. The primary responsibility of the database administrator is to organize a well-defined database for the organization.

Better Security: The traditional file system develops applications temporarily. The centralized approach of the database system provides various security constraints, which further improve the overall system's security.

- **Data backups:** DBMS technology allows the data to be backed up efficiently. The data backup can happen from a single location very simply.
- **Applications customisation:** The applications can be customised according to the customer's needs without tailoring the entire database.
- **Simple data management:** It provides robust tools for users to enter, delete, change or export corporate data.
- **Data durability:** DBMS ensures data durability. In the mishaps such as power outages or any other disaster, the data in the system remains unaffected.
- **Make sure to elaborate with at least one sentence while answering the DBMS interview questions like advantages, features, etc.**

Question 3: What Is “Redo” In a Database?

“REDO” in the database is used for rollback commands. When a database transaction takes place, it goes to the database buffer cache. Then, the data is coded to “REDO” logs from the “REDO” buffer. The data recovery operations need these logs for future use. The recovery of information is not possible without these logs. The rollback command clears the undo data.

The Redo entries record the data, which the users can utilise to reconstruct all changes made to the database. The redo log also protects the rollback data.

Question 4: Explain the Difference between “GROUP BY” and “ORDER BY”.

S.No.	GROUP BY	ORDER BY
1.	The GROUP BY command is used to combined data	The ORDER BY operation is used for sorting data
2.	It is utilized to form summaries in the result section	It is utilized for sorting results by a list of expressions
3.	The SQL clause specifying with the SELECT statement to retrieve data, gets grouped according to the one or more column.	The SQL clause specifying with the SELECT statement to return a result set, where the rows gets sorted according to a specific order.
4.	Helps to combine the data depending upon one or more columns.	Helps to organise the retrieved data in either ascending or descending order.
5.	It is used to group the rows having the same value.	Sort the results either in ascending or descending order.
6.	Could be allowed to CREATE VIEW statement.	Not used in the CREATE VIEW statement.
7.	In select statements, GROUP BY is always used before the order by keyword.	In select statements, ORDER BY is used is always used after the group by keyword.
8.	There is no possibility of keeping the attribute in the GROUP BY statement under the aggregate function.	Attribute can be under the aggregate function.

Question 5: While Creating Views, What Constraints Can You Apply?

The lists of constraints that can be applied are described below:

- The definitions of a complete text index cannot be applied.
- The views are only for the current database.
- You cannot create temporary views.
- The default definitions are not associated.
- Integrity constraints decide the functionality of delete and insert command.
- Triggers are associated with views.

Question 6: Explain All Categories of Database Languages.

The four categories of database languages are listed as below:

- **Data Manipulation Language:** Data manipulation languages or DML are used to make changes in updated information. The commands for data manipulation languages include update command, select command, delete command, insert command, and many more. DML commands handle the stored information in the database. These commands are not permanent. This means rollback operation is possible in DML commands.

As the name suggests, it enables the users to access and manipulate the data that is stored in the existing schema objects. The mentioned commands deal with the user requests for all data modification types. DML command dealing with data retrieval is known as Data Query Language.

Some of the DML commands are mentioned below-

- **SELECT:** It cannot manipulate the data, rather only access it.
- **INSERT:** It is used to insert the rows into the table.
- **UPDATE:** Helps to update the column/ table within a table.
- **DELETE:** It deletes the existing records from the table.
- **MERGE:** Merges the data from the source table to the targeted table.
- **CALL:** It is used to execute the Structured Query Language (SQL) from within SQL.
- **Data Definition Language:** Data definition languages or DDL are used to define and update the data. The commands for data definition languages include alter, create, truncate, drop, rename, and many more. The commands of data definition languages can store shared information.

It is a set of special commands that allows the users to define and modify the structure and the metadata of the database. Its commands can alter the structure of the database. Each change that is implemented by the DDL command is auto-committed.

Some of the DDL commands are mentioned below-

- **CREATE-** Creates the database or its schema objects.
- **DROP:** Deletes the database or its schema objects.
- **ALTER:** Modifies the structure of the database objects.
- **TRUNCATE-** Removes the whole content of the table along with deallocation without affecting the structure.
- **COMMENT:** Adds comments about the views, tables, and columns into the data dictionary.
- **RENAME:** Helps in changing the name of the existing table or a database object.
- **Transaction Control Language:** Transaction control languages or TCL are used for handling database transactions. The commands for transaction control languages consist of rollback, commit, save point, and many more. Transaction languages are mainly used for handling the modifications made by data manipulation languages.
- They are the special commands that deal with transactions within a database. The transactions play an important role for the execution of different tasks within a database. They are a collection of different tasks that are treated as a single execution unit by DBMS.

Some of the TCL commands are mentioned below-

- **COMMIT:** It is used to permanently save all the modifications that are done by the DML commands.
- **ROLLBACK:** It undoes the transactions that have not already been permanently saved or committed.
- **SAVEPOINT:** Creates a point within the groups of transactions. It does so to either save or roll back later.
- **AUTOCOMMIT:** Either enable/ disable the auto-commit process that commits each transaction after its execution.
- **Data Control Language:** Data control languages or DCL are used to take away or give database access to a user. The commands data control languages are “revoke” and “grant.” The syntax of the data control language is similar to the coding language. Data control language is related to security concerns. The GRANT DCL command provides access to the user. The REVOKE DCL command is used to withdraw the access privileges.

Some of the DCL commands are mentioned below:

- **GRANT:** Provides user access to the database or its objects.

- REVOKE: Revokes user access to the database system.

This is one of the DBMS important interview questions; collect your thoughts before answering this question. It will help you to answer smoothly without appearing flustered.

Question 8: Define the Nested Loop with an Example.

It is a loop over a loop. It includes some entries of the outermost loop within an internal loop. The nested loop follows three steps. First is the identification of the outermost table. Second is allocating the internal table to the outermost table. Then, for every outermost table row, retrieve the internal table rows. Let's consider an example.

Select row4.*, row5.* from rows, row5 where rows.row 4 =row 5.row 5;

The processing of the above query will take place in the below method:

- For I loop (select *from row4)
- For J loop (select * from row5 where row5=I.row 4)
- Loop end;
- Loop end;

Features of the nested loop-

- For loop can be put inside another for a loop.
- It is a loop within a loop.
- Any number of loops can be defined inside another loop.
- The nesting level can be defined n times.
- Any type of loop can be defined inside another loop.
- A while loop can be defined inside a for loop.

The main advantage of the nested loop is that it can help in quick retrieval of the first few rows of the result set. The process can be done without having to wait for the entire result to be determined.

DBMS interview questions and answers like this should be answered keeping the holistic view in mind. Make sure to mention the features, advantages or both. It adds depth to your answer. It makes for the most important DBMS interview questions. Make sure not to miss this topic while preparing.

Question 10: Explain the difference between the “TRUNCATE” and the “DELETE” Command.

S.No.	TRUNCATE Command	DELETE Command
1.	This command removes all the table rows	This command deletes only specific rows of the table
2.	This command doesn't keep a log	This command handles a log

- | | |
|---|---|
| 3. Truncate cannot be a rollback | The delete can be a rollback operation |
| 4. Truncate is a fast command | Delete is a slow command |
| | Delete command uses row log |
| 5. The truncate command uses table log | |
| 6. Marks the extents of a table for deallocation. | Removes one or more records from a table inside the database. |
| 7. Not possible to use where clause. | Possible to use where clause. |
| 8. Cannot use indexed views. | Can use indexed views. |
| 9. Reset the high level watermark. | Won't reset the high level watermark. |
| 10. No triggers will get fired. | Triggers will get fired. |

3. What is a schema?

The schema of a database refers to the skeletal structure of the logical view of a database. It defines the relations between the different cells, rows, columns and tables of a database. It also specifies all the constraints and details of different cells in a database. It also defines the relations shared between the various cells, rows, columns and tables of a database. A schema is a conceptual model or notion that aids in the organization and interpretation of data. It is helpful because they enable us to take shortcuts when comprehending the large quantity of data available in our surroundings.

4. What are the three levels of database?

The three levels of database include- a) Physical b) Conceptual c) External

5. What are the three users of DBMS?

The three main classifications of DBMS users include- a) End users b) Application Programmers c) Database Administrators

6. What are the four major uses of DBMS?

Four major uses of DBMS include: 1) Data retrieval 2) Data indexing 3) Data integrity 4) Data redundancy

8. What are the five types of databases?

Types of databases are mentioned below- 1) Hierarchical 2) NoSQL 3) Network 4) Object-Oriented 5) Relational

